



Research paper

Investigating grid-connected green power systems' energy storage solutions in the event of frequent blackouts

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ABSTRACT

When it comes to dealing with power outages, hybrid grid-connected systems seem like a promising option, especially for underdeveloped nations. For supplying power during periods of unpredictability, a PV/battery system connected to the grid is considered in this study. Batteries were used as a backup system to compensate for main grid outages in this paper, and five distinct types of energy storage battery technologies were compared: lead-acid battery (LA), lithium-ion battery (LI), vanadium redox battery (VR), nickel-iron battery (NI), and zinc-bromine flow battery (ZBF). The study was conducted in El Dabaa, Matrouh Governorate, Egypt (31° 1.6'N, 28° 26.6'E) using the Hybrid Optimization Model for Electric Renewables (HOMER) software. The investment cost, the renewable energy friction, and the excess electricity are the main comparative indexes considered. In the absence of outages, the grid is totally dependent on, with a total system NPC of \$58,000 and a COE of \$0.089/kWh for a short-term investment, \$81,883 and \$0.089/kWh for a medium-term investment, and \$121,382 and \$0.089/kWh for a long-term investment. The study reveals that the investment plan's duration had no impact on the proposed solar PV/battery system size, renewable energy percentage, CO₂ emissions, or unmet electric load.

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1. Introduction

With the world's population increasing in lockstep with technological and industrial advancements, as well as the increasing demand for energy supplies in various industries to meet the population's needs, the use of renewable energies has become unavoidable, even more so given the expected depletion of traditional energy sources.

Solar energy has a unique niche among renewable energy sources. Since the year 2000, the total solar photovoltaic capacity of the globe has constantly expanded. The overall global capacity for PV systems is estimated to reach 600 GW by 2019 (Victoria et al., 2021).

Recurring power outages in developing nations have a profound influence on the lives of people in these countries' socioeconomic and political contexts. One of the most intriguing solutions is to use solar battery systems to overcome this problem. Considering the high expenses of supply and establishment for PV-Battery systems, it is necessary to estimate the precise power consumption and, as a result, the right side of the PV panel and battery energy storage system to prevent oversizing and to meet energy needs at the lowest feasible price.

Barakat et al. (2020) state that the primary criteria for assessing the performance of grid-connected hybrid systems are the system's cost, reliability, and greenhouse gas emissions reduction. Numerous studies have shown the usefulness and performance of the hybrid grid-connected system in resolving the issue of energy outages in several locations worldwide. Several instances are shown below. When a hybrid-grid-connected system is used, the main grid's power charges are decreased. Najafi Ashtiani et al. (2020) showed this by proposing a solar PV and battery system that is connected to the utility. Their findings indicate that the suggested solution might lower grid power prices by up to 16.8 percent. Similarly, Samy et al. (2021a) presented a grid-connected hybrid PV/wind/fuel cell system employing the hybrid search optimization approach to provide energy during times of electrical outage. Egypt's grid electricity bills have dropped by 37.2% because of the findings.

Farquharson et al. (2018) found that certain hybrid grid-connected systems could not outperform the grid system economically. Using a backup diesel generator system only raises the long-term investment cost of the system, according to their research; this is compared to using the grid exclusively.

Power plays a larger role in the Sustainable Development Scenario, accounting for 31 percent of total electricity consumption (World Energy Outlook, 2019).

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Nomenclature

N_{PV}	Number of PV modules
D_f	PV derating factor
G	Global irradiance incident on the titled plane (kW/m^2)
G_{ref}	Solar radiation at reference conditions (1 kW/m^2)
K_T	Maximum power temperature coefficients
P_{PVR}	Module output power (kW)
T_{amb}	Ambient temperature ($^{\circ}\text{C}$)
T_c	PV cell temperature ($^{\circ}\text{C}$)

Abbreviations

P_{PV}	PV produced power (kW)
IRR	Internal rate-of-return
LI	Lithium-Ion Battery
MGFF	Main grid failure frequency
NI	Nickel–Iron Battery
NOCT	Normal operating cell temperature
NPC	Net Present Cost
NREL	National Renewable Energy Laboratory
O&M	Operating and maintenance
PV	Photovoltaics
RE	Renewable Energy
LA	Lead-Acid Battery
COE	Cost of Energy
DG	Diesel Generator
ESSs	Energy Storage Systems
HOMER	Hybrid Optimization Model for Electric Renewables
SDGs	Sustainable Development Objectives
SPB	Simple payback period
VR	Vanadium Redox Battery
WT	Wind Turbine
ZBF	Zinc–Bromine Flow Battery

The Sustainable Development Objectives (SDGs) were approved by the United Nations General Assembly in September 2015. The SDGs include 17 goals, including eradicating poverty and hunger, enhancing health and education, fighting climate change, and preserving forests, among others (Nations, 2015). Multiple goals may be achieved by increasing the production of renewable energy (RE). Because of this, achieving the Sustainable Development Goals will lead to a substantial increase in the demand for renewable energy.

For three reasons, RE has significant synergistic impacts on the SDG. First and foremost, it gives energy, which serves as a foundation for making progress toward a variety of aims and objectives. Energy has the potential to increase the productivity of agriculture and, as a result, contribute to food security (Mush-taq et al., 2009). Because renewable energy enables nations to become self-sufficient in terms of fuel imports, it also helps to the aim of guaranteeing a stable and sustainable source of electricity (Tsagkari et al., 2021; Resniova and Ponomarenko, 2021).

Second, unlike some of the presently prevalent alternatives, such as conventional biomass and coal, renewable energy does not cause local environmental harm. It may therefore contribute to the aim of guaranteeing healthy lives by replacing the source

of indoor air pollution and improving local air pollution as a result of co-emitted air pollutants, both of which would be beneficial (West et al., 2013; Dinkelman, 2011; Part, 2014; Daka and Ballet, 2011).

And last, of all, renewable energy emits very little CO_2 and therefore has no impact on climate change. While combating climate change is one of the SDGs, it affects the long-term chances of accomplishing almost all the other SDGs (Schwerhoff and Sy, 2017). The effects of climate change are so broad and severe, as outlined in the IPCC report, that any progress toward the SDGs that is not sustainable may be short-lived (Part, 2014).

PV and diesel are widely employed as backup energy solutions during grid power outages, according to prior studies. Unmet energy demand is often met by using a diesel generator in developing countries. The use of solar PV as a backup system for electricity delivery during blackout times might be a feasible alternative to using diesel generators in places where there is a significant amount of solar energy available.

Because of the integration of Energy Storage Systems (ESSs) with renewable energy systems and the development of renewable energy micro-grids, it is now possible to use a greater percentage of renewable energy, opening the way for the creation of hybrid renewable energy systems (Barakat et al., 2020; Samy et al., 2021b; Al-Ghussain et al., 2020; Azerefeign et al., 2019; Tazay et al., 2020; Samy et al., 2020; Samy and Barakat, 2019; Samy S. Barakat, 2019).

A few decades ago, among other technologies, lead-acid batteries were the most frequently utilized battery energy storage systems for electric power system applications. Lithium-ion batteries, on the other hand, have recently grown in popularity because of their lower maintenance requirements, improved safety, higher gravimetric energy densities, and unique volumetric properties. In comparison to lead-acid batteries, these batteries are more compact and lighter in weight. Many additional battery energy storage technologies, such as vanadium redox battery, ZBF battery, Ni–Cadmium battery, and sodium–sulfur battery, are also used for energy storage (Jitson and Chepanich, 2021; Xiao et al., 2021; Perez Castillo et al., 2018; Jaiswal, 2017; Skyllas-Kazacos and McCann, 2015).

Table 1 provides a comprehensive review of hybrid systems that use various battery technologies. Only a few researchers, however, have concentrated on an in-depth comparative performance analysis of off-grid hybrid systems using different battery technologies. The current research demonstrated this by examining the performance of a PV/wind/diesel generator, as well as the sensitivity analysis of the optimal system.

To address the issue of electrical blackouts, the authors of this study propose an optimum design and comparison of many feasible options for hybrid grid-connected energy supply systems. In this research, five battery technologies are compared: lead-acid battery (LA), lithium-ion battery (LI), vanadium redox battery (VR), nickel–iron battery (NI), and zinc–bromine flow battery (ZBF). The current study intends to close various research gaps that have been identified, which are as follows:

- Diesel generators are the principal source of electricity in third-world nations during grid outages. Numerous alternative energy systems for diesel generators may be presented and evaluated to aid in making an informed choice about how to resolve this issue.
- Analysis of different battery technologies in a hybrid operation of the grid, PV, and battery technology with a minimum COE, the highest internal rate-of-return (IRR), the lowest simple payback period (SPB), minimal pollutant emissions, and high renewable penetration is the goal of this study.

Table 1

A thorough review of several hybrid system arrangements using a diversity of battery technologies.

Ref.	System configuration	Location	Battery technology	COE	Findings
Li et al. (2020)	Off-grid WT/DG/Battery	China	LA, ZBF, LI	0.471 to 2.34 \$/kWh	The DG/ZB system has the lowest NPC and COE. DG/LI is also determined to be the most eco-friendly system compared to DG/ZB and DG/LA.
He et al. (2021)	Off-grid WT/PV	Pakistan	battery, thermal energy storage, pumped hydro storage, and hydrogen storage	0.1224 to 0.225 \$/kWh	Thermal energy storage is usually the least expensive option.
Kebede et al. (2021)	Photovoltaic Grid-Connected System	Ethiopia	LA, and LI	0.32 to 0.34 €/kWh	Under the specified specs and load profile, lithium-ion batteries are more economically feasible than lead-acid batteries.
Kumar et al. (2021)	Off-grid PV/diesel generator	India	LA, LI, VR, and ZBF	\$0.167/kWh and \$0.155/kWh	Zinc-bromine flow batteries provide the best rate of return on investment and renewable penetration while requiring the shortest simple payback time and emitting the least pollutants.
Eteiba et al. (2018)	Off-grid PV/Biomass generator	Egypt	Flooded lead-acid (FLA), Lithium Ferro Phosphate (LFP), and Nickel-Iron (Ni-Fe)	\$(0.0834 to 0.0837)/kWh	Nickel Iron (Ni-Fe) batteries are more cost-effective than the other two battery technologies within the given specifications and load profile.
Escalante Soberanis et al. (2018)	Off-grid PV/WT/diesel generator	Canada	Zinc air flow battery	\$(0.745–1.2)/kWh	The battery system delivers power during low demand times, while the DG runs continuously during high demand periods. Combining a battery system with renewable energy sources could reduce fuel usage by up to 75%.
Merei et al. (2013)	Off-grid PV/WT/diesel generator	Aachen, Germany and Quneitra, Syria	LA, LI and VR	\$(0.41–1.98)/kWh	Using batteries in conjunction with renewable energy is both cost-effective and environmentally friendly. However, combining redox-flow batteries with renewables is optimal.

- This research investigates the impact of several investment plans (short, medium, and long term) on the system's cost by tracking how the cost of the system changes over time.
- The present case study, which may be applied to various scenarios, may assist in decision-making about the most cost-effective, reliable, and environmentally friendly hybrid grid-connected system operating during power outages.

The paper is arranged as follows: the first section outlines the study subject, discusses the issue's background, and reviews previous studies. Methodology and modeling are covered in Section 2. Section 3 summarizes the results and discusses the process. Finally, Section 4 presents the conclusion.

2. Methodology and modeling

The HOMER Pro[®] software was used to analyze and evaluate the proposed method for dealing with the problem of frequent power outages, as well as to evaluate different battery technologies (Lambert et al., 2006). Input load data, technical specifications, and financial characteristics of various system components form the basis of the current system design. Different battery technologies are used to run the system to achieve the rank-based optimum system response for the considered location in HOMER with the least amount of NPC and COE.

It is the goal of this research to develop an overarching methodological and simple framework for the optimal design and operation of a solar PV/battery system connected to a grid that experiences frequent power outages, to make the planning and use of such systems more convenient and efficient. It is separated into three different parts: (1) an initial assessment, (2) a study of design optimization, and (3) results. The technique mentioned in this study is summarized in Fig. 1, which is shown below.

As part of the first stage, pre-optimality analysis, the available renewable energy sources, as well as the, predicted locally based load demand are thoroughly investigated. Current energy sources and metrological characteristics have restrictions that must be addressed. Following the completion of this phase, system designers will be better prepared to identify viable energy technologies and construct an ideal mix of those technologies to maintain a balanced load demand.

In combination with the HOMER Pro[®] program, the selected energy technologies are appropriately modeled and analyzed in the context of a total of five distinct battery technologies. While meeting electric load demand, we want to cut net present cost, energy cost, and pollution as much as possible. While keeping in mind design objectives, the least costly and ecologically friendly grid-connected system with the best level of dependability is selected and assessed in terms of numerous operational, financial, and environmental criteria during the third phase's review.

2.1. The research area's presentation

The study area is a small primary school that is in El Dabaa, Matrouh Governorate, Egypt at 31° 1.6' N latitude and 28° 26.6' E longitude. Due to its location within the region characterized by a tropical desert climate characterized by drought and prolonged sunlight, even during the winter season, Matrouh Governorate is one of the desert border governorates. The region is also exposed to a high amount of solar radiation due to its location within the region of the desert border governorates.

On the map below (Fig. 2), you can see the geographic location of El Dabaa, which was selected as the research region for this study. The data on irradiance and ambient temperature are provided in Figs. 3 and 4, respectively.

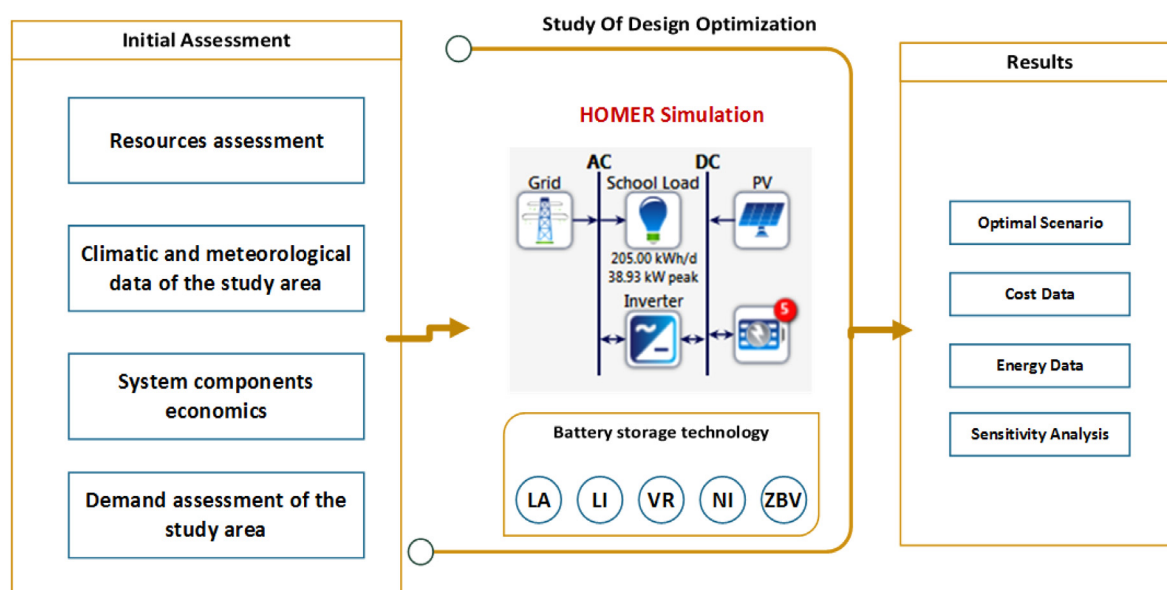


Fig. 1. Diagrammatic representation of the study techniques.



Fig. 2. Egypt map for the geographical location of the studied site.

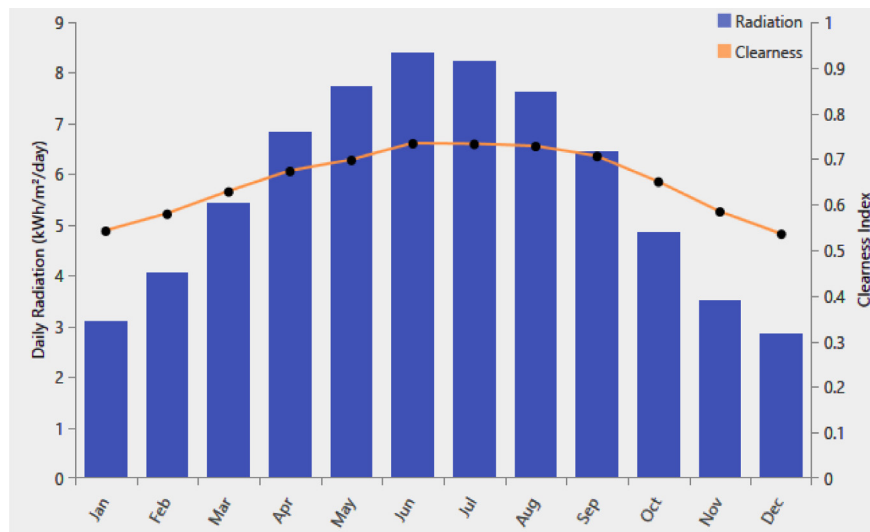


Fig. 3. The monthly average solar irradiance data of the study site.

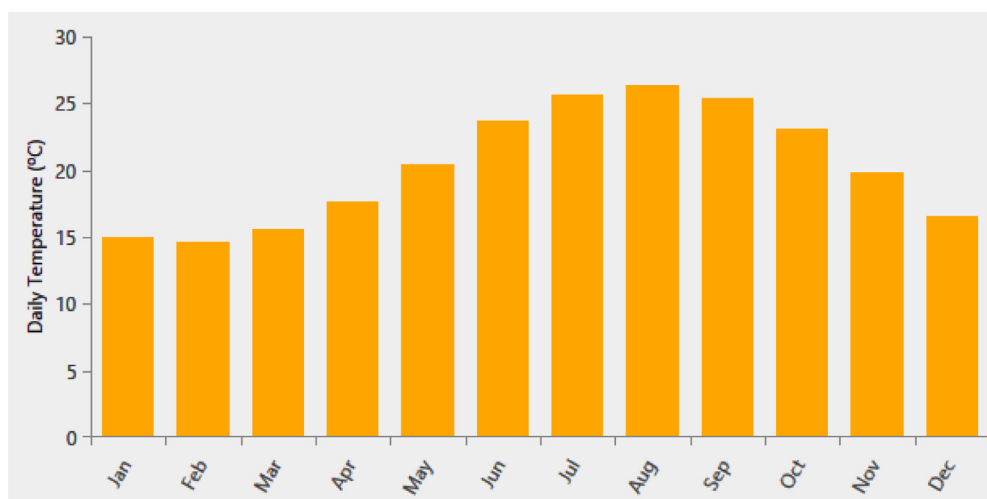


Fig. 4. The monthly average temperature data of the study site.

2.2. System's component modeling

When it comes to hybrid power systems, the National Renewable Energy Laboratory's (NREL) HOMER Pro simulation program is the most often utilized tool. The HOMER simulation makes use of inputs such as data on resources, component technical and economic factors, load data, and other project parameters. It performs a variety of system configuration simulations before settling on the most cost-effective option.

Fig. 5 illustrates a grid-connected hybrid system that utilizes a photovoltaic (PV) array, a bidirectional converter, and a battery bank (LA, LI, VR, NI, and ZBV) to power a small school.

2.2.1. The grid modeling

The main grid energy supply system is modeled based on the load profile of the users in the proposed study approach. Fig. 6 shows the electrical energy profile of a small school, which fluctuates depending on the season. It depends on the month of the year since during school months, the school's consumption is higher than during summer break.

205 kWh/day on average, with a 15 percent day-to-day random fluctuation, with a peak demand of 39 kW for the specified location is the average daily electrical energy consumption. To conduct economic analysis, a nominal discount rate of 8.75% and an inflation rate of 6% were considered.

HOMER provides the ability to identify grid failures to represent an unstable grid. It enables planned outages to be arranged according to the time of day and month of the year. It enables the generation of random outages throughout the year by setting parameters. Outages are depicted as a series of time steps during which no power may be acquired from or sold to the grid are shown. As a result, HOMER creates a time series of outages and presents them in a Grid outages DMap by utilizing input parameters. There is black when the grid is down, and green when it is running normally. All simulations utilize the grid outage time data shown in Fig. 7.

As part of this research, the main grid failure frequency (MGFF), which is the number of times the grid will fail each year, is addressed as a sensitivity variable, with a 100% variability in

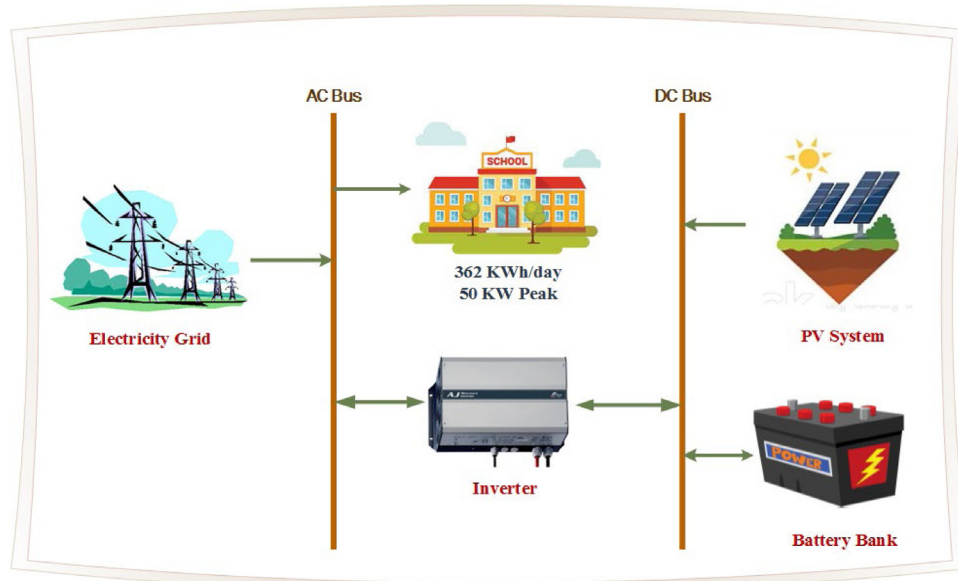


Fig. 5. The proposed system's architecture.

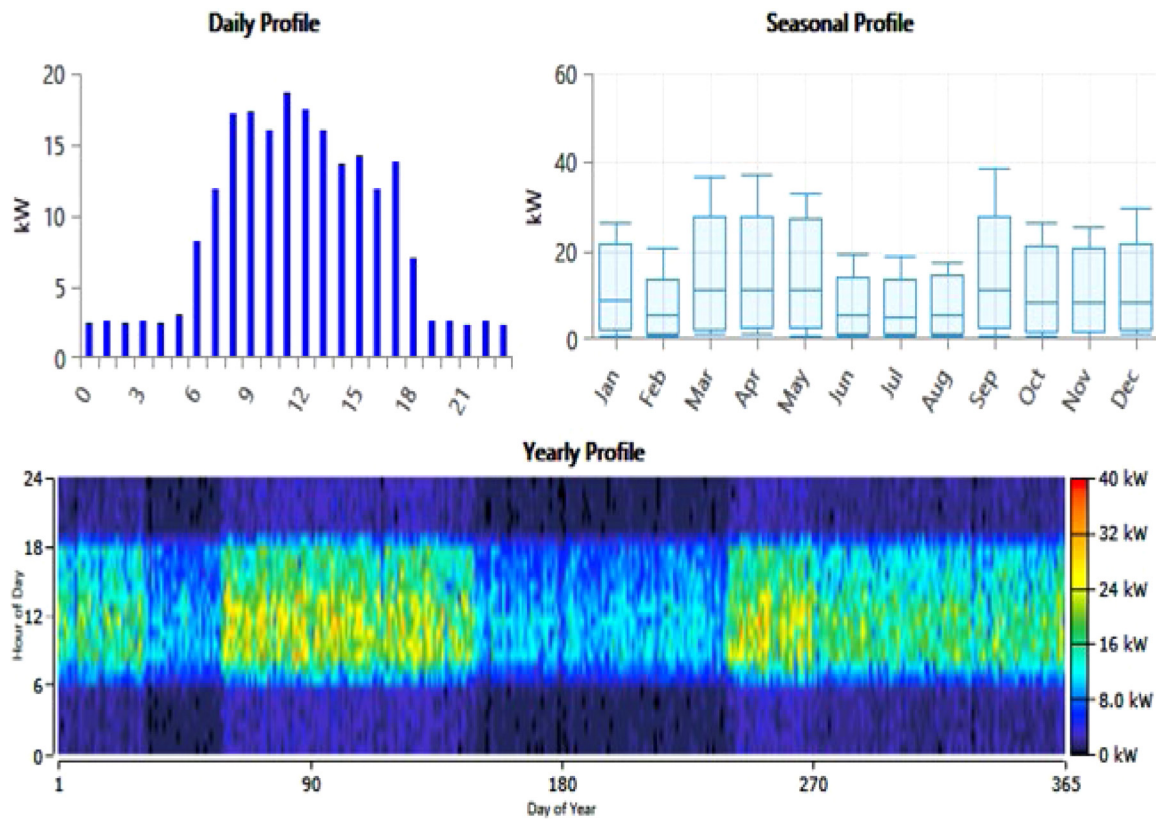


Fig. 6. The studied area load profile.

repair time. Table 2 provides a list of the simulated MGFFs, as well as their relationship to the Grid's availability.

2.2.2. PV generator modeling

The electrical energy produced by the PV system is calculated via eq (1) (Duffie and Beckman, 2013).

$$P_{pv} = (P_{PVR} \times N_{pv} \times DF_{pv}) \left(\frac{G(t)}{G_{base}} \right) (1 + K_T (T_c - T_{base})) \quad (1)$$

Where:

$$T_c = T_{amb} + G \left(\frac{NOCT - 20}{G_{NOCT}} \right) \quad (2)$$

With a 25-year lifetime assumed, the capital and replacement expenses for the PV array are \$1,187, while the operating and maintenance (O&M) costs are assumed to be \$5.

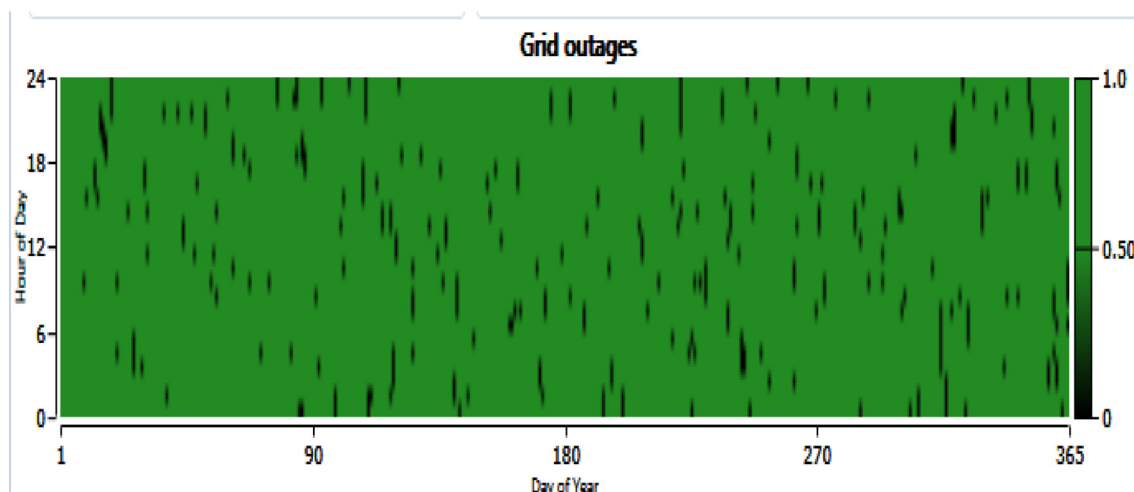


Fig. 7. The Grid outages DMap.

Table 2

The relationship between the Grid availability and the simulated MGFF.

MGFF (1/yr)	Grid availability
0	100%
250	97%
500	94%
750	91%
1000	89%
2000	77%
3000	66%

2.2.3. The battery modeling

To determine which type of battery is most applicable in hybrid systems, five different types of batteries were tested, including the Hoppecke 24 OPzS 3000-Vented lead-acid (LA), the Blue Ion 2.0 lithium-ion battery (LI), the redT vanadium redox battery (VR), the Iron Edison nickel-iron battery (NI), and the Redflow ZBM2 zinc-bromine flow battery (ZBF). Table 3 contains the important details about the batteries that were chosen.

2.2.4. Converter

To maintain the energy flow between the direct and alternating current components, a Studer AJ 2400–24 inverter has been used. The inverter's efficiency is to be about 94%, and it has a ten-year life expectancy. The rectifier has a capacity of 100% and an efficiency of 94%. In the case of the converter under consideration, the capital, replacement, and operation and maintenance expenses are \$725/kW, \$650/kW, and 10 \$/kW/yr, respectively.

3. Results and discussion

The goal of this research is to identify the most efficient and cost-effective strategy for addressing the issue of electrical outages in the future. HOMER simulates all potential system configurations for a small primary school in Egypt, even some that are not feasible. In this article, the influence of the project's longevity on the optimal system design was investigated in further detail. Specifically, simulations were run for short-term investments (10 years), medium-term investments (15 years), and long-term investments (30 years). Results of the simulation are reported in Tables 4–6.

The data in Tables 4–6 demonstrate that, in the absence of interruptions, the grid is completely reliant on to meet the required load, with a total system NPC of \$58,000 and a COE of \$0.089/kWh

in the case of a short-term investment, a total system NPC of \$81,883 and a COE of \$0.089/kWh in the case of a medium-term investment, whereas in the case of a long-term investment the total system NPC is \$121,382 and the COE is \$0.089/kWh.

Nevertheless, it should be noted that when the MGFF was raised to 250, the grid-connected PV/ ZBF system was selected as the optimal system and that the number of batteries was always four, regardless of the kind of investment made. Short-term investments resulted in a total NPC of \$71,556 and COE of \$0.088/kWh, while medium-term investments resulted in a total NPC of \$97,880 and COE of \$0.082/kWh, and long-term investments result in a total NPC of \$137,708 and COE of \$0.078/kWh.

In addition, it should be highlighted that, as the investment term lengthens, the NPC value increases while the COE value decreases.

Meanwhile, at MGFF rates of 500, 750, 1000, and 3000, it was determined that the grid-connected PV/NI system was the most efficient. As shown by the data, the number of batteries was fixed at 40 in the cases of MGFF of 500, 750, and 1000, but the number grew to 80 in the case of MGFF of 3000, although the size of the solar PV system changed as the MGFF went from 500 to 3000 according to all three investment instances examined. There is an increase in the overall cost of the system in all three investment scenarios analyzed because a larger solar photovoltaic system was installed.

However, in the case of the MGFF of 2000, the grid-connected PV/ LI system was determined to be the most optimal system, consisting of three batteries and a PV of 52.9 kW in the case of short-term investment, three batteries, and a PV of 52.8 kW in the case of medium-term investment, and three batteries and a PV of 53.4 kW in the case of long-term investment.

Fig. 8 demonstrates the relationship between the NPC, the COE, and the MGFF. When the grid outage frequency is between 250 and 1000, the total cost of a hybrid system increases slightly, whereas when the grid outage frequency is between 2000 and 3000, the total cost of a hybrid system increases significantly faster. It is more expensive to install a larger solar PV array in a hybrid system to compensate for grid interruptions.

Fig. 9 illustrates the correlation between the share of renewable energy in the total energy mix and the frequency of power system outages. It is evident that in the absence of interruptions, the system is completely reliant on the grid to provide the demand, and as a result, the renewable energy fraction was zero. While outages were occurring, it can be noticed that the value of the renewable energy factors varied from 72.0 percent in the

Table 3
Specifications of the introduced batteries.

Parameter	Lead-Acid Battery	Lithium-Ion Battery	Vanadium Redox Battery	Nickel-Iron Battery	Zinc-Bromine Flow Battery
Nominal voltage (V)	2	51.2	48	1.2	48
Max. capacity (Ah)	3570	328	417	1000	215
Nominal capacity (kWh)	7.15	16.8	20	1.2	10.3
Max. charge current (A)	610	263	105	500	52.1
Max. discharge current (A)	610	328	105	500	104
Roundtrip efficiency (%)	86	97	75	85	75
Capital cost (\$)	722	15,000	10,700	970	8,000
Replacement cost (\$)	665	13,800	9,300	890	7,400
O&M cost (\$/year)	180	1	1	1	1
Lifetime (years)	15	25	25	20	25

Table 4
A comparison of short-term (10-year) investments in optimum grid-connected PV/battery hybrid systems.

MGFF (1/yr)	Architecture	PV (kW)	Battery bank (No.)	Grid (kW)	Inverter (kW)	NPC (\$)	COE (\$/kWh)	Ren Frac (%)	Autonomy (hr.)
0	Grid only	-	-	300	-	58,001	0.089	0.0	-
250	PV/ZBF	43.8	4	300	26.6	71,556	0.088	72.0	4.82
	PV/NI	41.3	40		25.5	72,341	0.091	70.4	5.06
	PV/LI	42.1	2		26.0	77,085	0.096	71.1	3.89
	PV/VR	28.9	4		27.1	90,165	0.127	57.5	9.37
	PV/LA	41.0	24		25.1	110,293	0.140	70.5	14.06
500	PV/NI	43.3	40	300	27.4	74,253	0.092	72.1	5.06
	PV/ZBF	50.1	4		30.3	75,705	0.088	75.9	4.82
	PV/LI	45.6	2		27.8	78,233	0.095	73.6	3.89
	PV/VR	32.4	5		27.5	95,886	0.132	61.8	11.72
	PV/LA	43.1	24		26.9	111,221	0.139	72.7	14.06
750	PV/NI	46.7	40	300	28.1	75,862	0.093	74.0	5.06
	PV/LI	48.4	2		29.9	79,611	0.095	75.4	3.89
	PV/ZBF	47.8	5		28.4	80,379	0.097	74.1	6.03
	PV/VR	33.6	5		28.1	96,381	0.133	62.9	11.72
	PV/LA	43.5	24		27.4	111,740	0.141	73.3	14.06
1000	PV/NI	49.3	40	300	29.8	77,822	0.093	75.5	5.06
	PV/LI	52.2	2		31.6	81,224	0.094	77.2	3.89
	PV/ZBF	50.6	5		30.1	83,515	0.099	75.5	6.03
	PV/VR	36.1	5		29.6	97,268	0.132	65.4	11.72
	PV/LA	46.1	24		29.0	112,833	0.140	75.4	14.06
2000	PV/LI	52.9	3	300	32.7	93,023	0.113	77.2	5.84
	PV/NI	48.6	80		31.4	94,992	0.120	75.4	10.12
	PV/ZBF	58.4	6		32.9	98,975	0.117	77.3	7.24
	PV/VR	42.8	5		34.1	101,691	0.134	69.8	11.72
	PV/LA	47.3	24		31.5	116,182	0.149	76.5	14.06
3000	PV/NI	52.3	80	300	35.4	103,242	0.132	76.3	10.12
	PV/LI	45.9	5		36.5	114,215	0.152	74.2	9.74
	PV/ZBF	58.9	8		36.8	117,606	0.144	76.4	9.65
	PV/LA	46.1	24		34.0	119,963	0.162	75.2	14.06
	PV/VR	45.6	7		39.5	120,987	0.163	70.4	16.40

case of 250 MGFF and short-term investment to 77.4 percent in the case of 2000 MGFF and long-term investment.

Fig. 10 depicts the association between the size of the photovoltaic (PV) system and the frequency of grid outages. When considering the situation of 250 MGFF, it is worth noting that the size of the PV system increased from 43.8 kW to 47.8 kW and then 47.7 kW in the case of short-term, medium-term, and long-term investment, respectively. Whereas, in the instance of 500 MGFF, the size of the PV system was 43.3 kW, 43.8 kW, and 45.9 kW when considering short-term, medium-term, and long-term investments. It should be observed that when the MGFF was increased to 750 and 1000, the size of the PV system in the three investment situations under examination became quite close. MGFF values of 2000 and 3000 were quite similar in terms

of their effect on the size of the PV system in the three investment situations under consideration.

In Fig. 11, the percentage of excess electricity from the optimal system for various MGFFs according to the various project investment plans is depicted, with the ratio increasing because of increasing the size of the PV system to compensate for grid outages, as the ratio increases from 0 percent in the zero-outage case to 13.9 percent in the case of MGFF 3000. However, it should be highlighted that modifying the investment period had only a little impact on the proportion of excess electricity generated.

According to the different project investment plans presented in Fig. 12, the CO₂ Emissions from an ideal system for various MGFFs are displayed. The grid (base scenario) seems to be the most polluting system (emitting 47,289 kgCO₂/year), according to

Table 5

A comparison of medium-term (15-year) investments in optimum grid-connected PV/battery hybrid systems.

MGFF (1/yr)	Architecture	PV (kW)	Battery bank (No.)	Grid (kW)	Inverter (kW)	NPC (\$)	COE (\$/kWh)	Ren Frac (%)	Autonomy (hr)
0	Grid only	-	-	300	-	81,883	0.089	0.0	-
250	PV/ZBF	47.8	4	300	27.6	97,880	0.082	75.0	4.82
	PV/NI	41.3	40		25.5	98,925	0.088	70.4	5.06
	PV/LI	43.4	2		25.9	105,197	0.092	71.8	3.89
	PV/VR	28.9	4		27.1	122,493	0.123	57.5	9.37
	PV/LA	41.0	24		25.1	152,582	0.137	70.5	14.06
500	PV/NI	43.8	40	300	27.3	101,411	0.089	72.4	5.06
	PV/ZBF	51.3	4		30.1	103,304	0.084	76.3	4.82
	PV/LI	44.9	2		27.8	106,655	0.092	73.2	3.89
	PV/VR	32.4	5		27.5	129,605	0.127	61.8	11.72
	PV/LA	44.0	24		27.0	153,761	0.135	73.3	14.06
750	PV/NI	47.0	40	300	28.0	103,534	0.089	74.1	5.06
	PV/LI	48.7	2		29.8	108,478	0.091	75.5	3.89
	PV/ZBF	48.3	5		28.4	109,579	0.094	74.4	6.03
	PV/VR	34.3	5		28.0	129,763	0.126	63.7	11.72
	PV/LA	44.4	24		27.5	154,491	0.137	73.9	14.06
1000	PV/NI	48.7	40	300	29.9	106,115	0.091	75.2	5.06
	PV/LI	52.3	2		31.6	110,593	0.091	77.2	3.89
	PV/ZBF	59.6	4		35.1	113,478	0.087	79.5	4.82
	PV/VR	36.1	5		29.6	131,394	0.126	65.4	11.72
	PV/LA	46.6	24		29.0	155,944	0.137	75.6	14.06
2000	PV/LI	52.8	3	300	32.8	126,353	0.108	77.2	5.84
	PV/NI	50.4	80		31.4	128,857	0.114	76.3	10.12
	PV/ZBF	58.4	6		32.9	134,319	0.112	77.3	7.24
	PV/VR	42.8	5		34.1	137,339	0.128	69.8	11.72
	Grid/LA	0.0	72		32.3	337,620	0.367	0.0	42.18
3000	PV/NI	52.4	80	300	35.5	140,071	0.127	76.3	10.12
	PV/LI	45.9	5		36.5	154,644	0.146	74.2	9.74
	PV/ZBF	58.9	8		36.8	159,268	0.138	76.4	9.65
	PV/VR	45.6	7		39.5	162,713	0.155	70.4	16.40
	PV/LA	49.0	24		33.9	165,724	0.155	77.1	14.06

the results. As seen in Fig. 12, the use of hybrid systems that are linked to the grid results in a large reduction in carbon dioxide emissions. The change in the investment plans had only a little impact on the quantity of carbon dioxide emissions produced. In the situations of MGFF ranging from 250 to 3000, it was discovered that the value of the emissions generated varied only little because of the convergence of the optimum systems that were chosen.

According to the different project investment plans represented in Fig. 13, the percentage of the unmet electric load of the ideal system for various MGFFs is depicted. It indicates that the grid (the basic scenario) has the least amount of unmet electric

capacity (with 0 percent). However, it should be noted that the rate of rising has been slow, with the maximum percentage not exceeding 0.1 percent. It is also worth noting that the change in investment plans had only a little effect on the unmet electric load ratios.

The internal rate of return (IRR percent) and the simple pay-back period (SPB) were utilized to determine the economic feasibility of the proposed system during the economic evaluation. The internal rate of return (IRR) is a measure of the project's genuine interest yield during its life. It is sometimes referred to as the rate of return on investment (ROI) or the rate of return adjusted for time. It is determined by computing the discount rate at which

Table 6

A comparison of long-term (25-year) investments in optimum grid-connected PV/battery hybrid systems.

MGFF (1/yr)	Architecture	PV (kW)	Battery bank (No.)	Grid (kW)	Inverter (kW)	NPC (\$)	COE (\$/kWh)	Ren Frac (%)	Autonomy (hr)
0	Grid only	-	-	300	-	121,382	0.089	0.0	-
250	PV/ZBF	47.7	4	300	27.7	137,708	0.078	74.9	4.82
	PV/NI	44.3	40		26.5	139,609	0.082	72.9	5.06
	PV/LI	43.3	2		26.0	147,386	0.087	71.8	3.89
	PV/VR	28.9	4		27.1	172,005	0.116	57.5	9.37
	PV/LA	41.9	24		25.1	220,838	0.133	71.1	14.06
500	PV/NI	45.9	40	300	27.3	142,908	0.083	73.5	5.06
	PV/ZBF	51.1	4		30.0	144,757	0.080	76.2	4.82
	PV/LI	45.5	2		27.8	149,229	0.087	73.6	3.89
	PV/VR	32.4	5		27.5	180,785	0.119	61.8	11.72
	PV/LA	45.9	24		27.0	222,264	0.130	74.2	14.06
750	PV/NI	47.6	40	300	28.0	145,743	0.085	74.4	5.06
	PV/LI	49.1	2		29.8	151,562	0.086	75.7	3.89
	PV/ZBF	50.0	5		28.1	153,466	0.088	75.0	6.03
	PV/VR	33.6	5		28.0	181,505	0.119	62.9	11.72
	PV/LA	46.6	24		27.4	223,246	0.132	75.0	14.06
1000	PV/NI	49.2	40	300	29.7	149,096	0.086	75.2	5.06
	PV/LI	52.6	2		31.5	154,256	0.085	77.2	3.89
	PV/ZBF	60.0	4		34.8	157,941	0.082	79.5	4.82
	PV/VR	36.1	5		29.6	183,007	0.118	65.4	11.72
	PV/LA	47.0	24		29.1	225,203	0.133	75.6	14.06
2000	PV/LI	53.4	3	300	32.6	175,831	0.102	77.4	5.84
	PV/NI	49.8	80		31.4	180,408	0.108	76.0	10.12
	PV/ZBF	56.0	6		32.6	187,832	0.107	76.4	7.24
	PV/VR	42.8	5		34.1	190,992	0.120	69.8	11.72
	Grid/LA	0.0	72		32.3	498,107	0.365	0.0	42.18
3000	PV/NI	52.1	80	300	35.5	196,051	0.120	76.2	10.12
	PV/LI	43.7	5		36.6	215,568	0.140	72.7	9.74
	PV/ZBF	60.0	8		36.4	224,828	0.131	76.7	9.65
	PV/VR	45.6	7		39.5	225,285	0.145	70.4	16.40
	PV/LA	49.4	24		33.9	239,216	0.151	77.3	14.06

the project's net present value equals zero. If the project's IRR is equal to or higher than the investor's needed rate of return, the project is likely to be judged acceptable. The project is usually rejected if it falls short of this criterion.

Simply said, the Payback Period (PBP) is the period required for an investment project to return its own original cost by the cash revenues generated by the project over which it is invested. This index is very important for private individuals and small businesses that may be cash strapped.

The variance of the IRR and SPB MGFFs as a function of the different project investment plans and MGFFs is shown in Fig. 14.

Base case (Grid alone) values of IRR and SPB are zero; however, for GMFFs of 2000 and 3000, the highest IRR and lowest simple payback period may be obtained. In addition, it is important to note that the change in investment plans had only a little impact on both IRR and the SPB.

4. Conclusions

An analysis of several investment plans has been carried out in this research, ranging from short-term (10 years) to medium-term (15 years) to long-term (25 years), as well as various battery



Fig. 8. The relationship between the NPC, COE, and the MGFF for (a) short-term, (b) medium-term, and (c) long-term investments.

technologies, to alleviate the issue of electrical blackouts. This investigation was carried out for a variety of grid outage frequencies (0 to 3000 interruptions per year).

For a school in El-Dabaa city, Egypt, the technical and economic feasibility of implementing a grid-connected solar/battery system with five distinct batteries was investigated using the

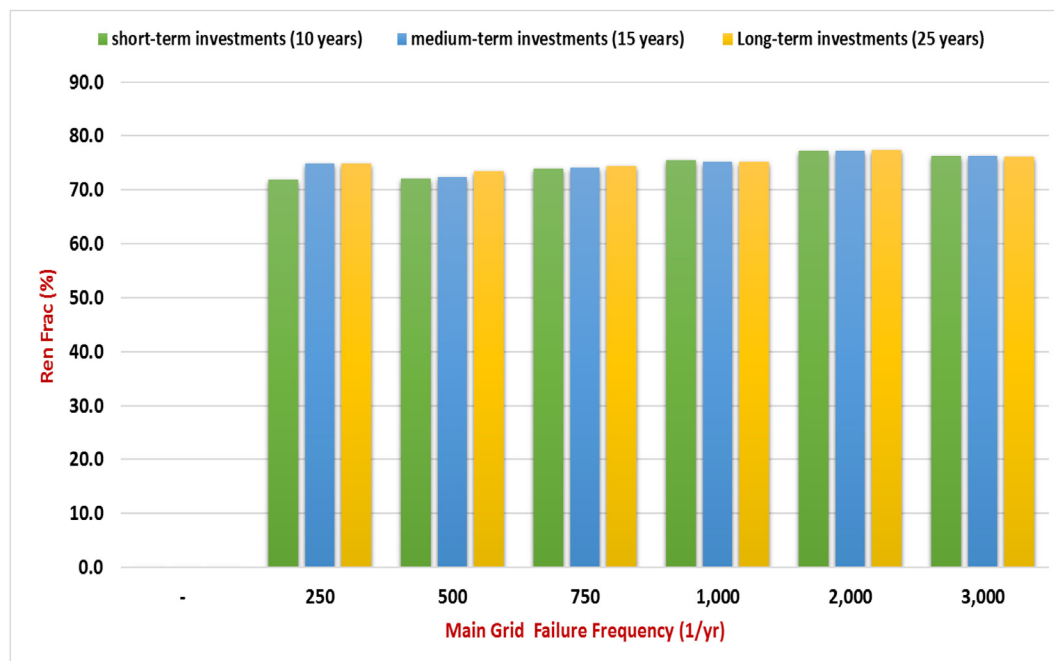


Fig. 9. The relationship between the renewable fraction NPC, COE, and the MGFF for short-term, medium-term, and long-term investments.

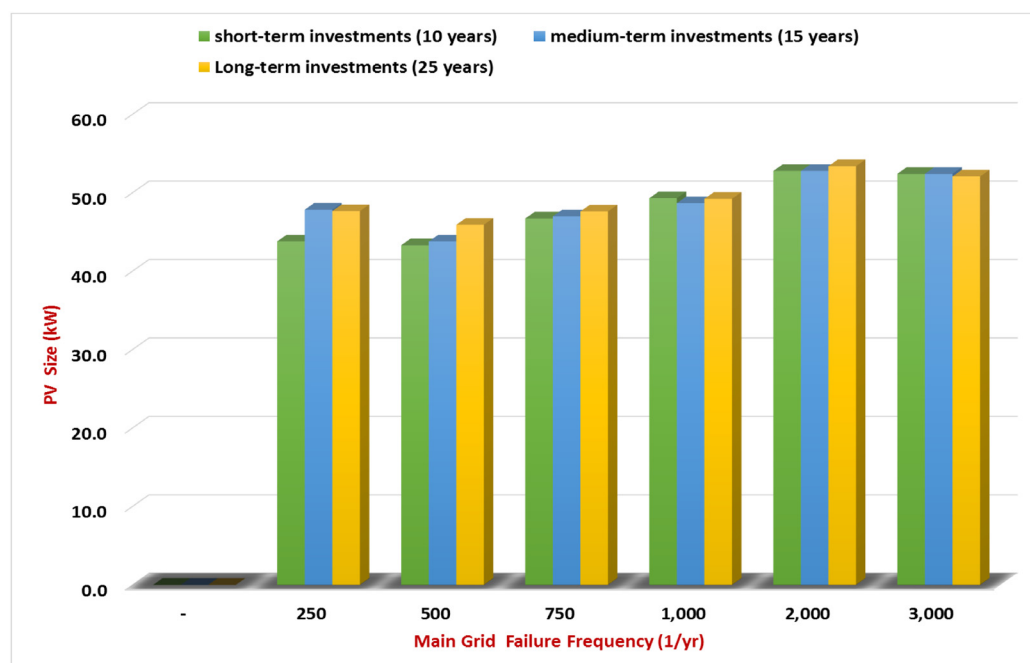


Fig. 10. The relationship between the PV size and the MGFFs for the three investment scenarios.

HOMER Pro software. The goal was to find a solution to the issue of frequent power outages.

The following are the most significant outcomes of the presented paper:

- The main grid is completely relied upon in the absence of interruptions, with a total system NPC of \$58,000 and a COE of \$0.089/kWh for a short-term investment, \$81,883 and \$0.089/kWh for a medium-term investment, and \$121,382 and \$0.089/kWh for a long-term investment.
- When the MGFF was increased to 250, the grid-connected PV/ZBF system was chosen as the best option, and the number of batteries was always four, independent of the investment type.
- While the investment period lengthens, the NPC value raises, and the COE value declines.
- During interruptions, the value of renewable energy fraction fluctuated from 72.0 percent for 250 MGFF and short-term investment to 77.4% for 2000 MGFF and long-term investment.

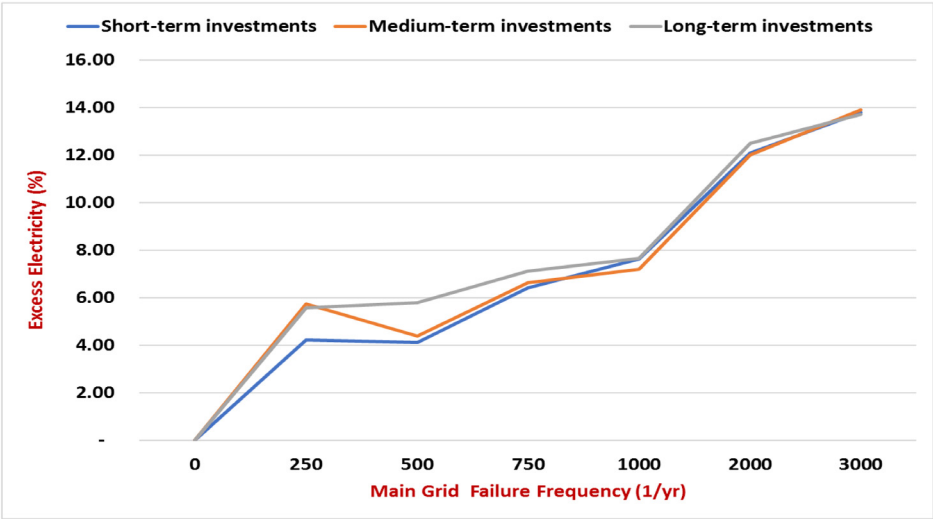


Fig. 11. Excess electricity for various MGFFs per the various project investment plans.

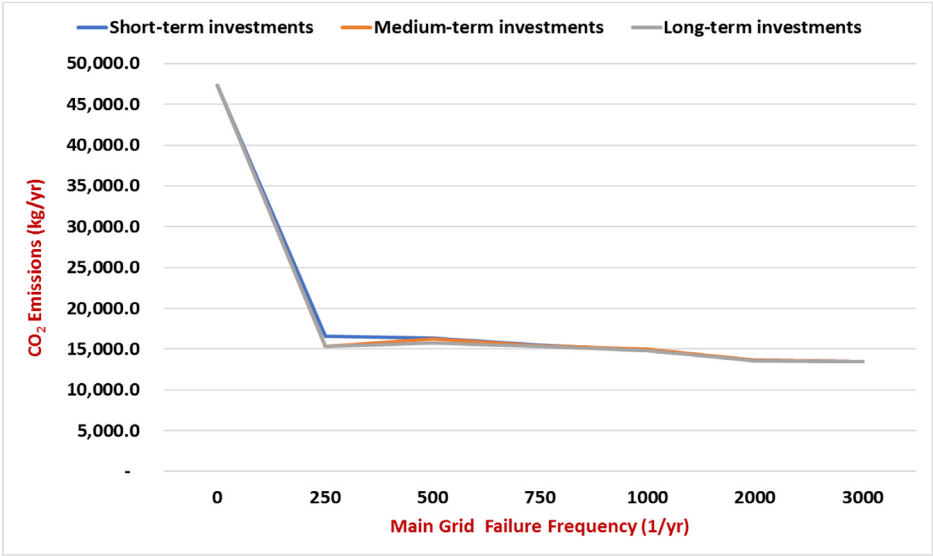


Fig. 12. CO₂ Emissions for various MGFFs per the various project investment plans.

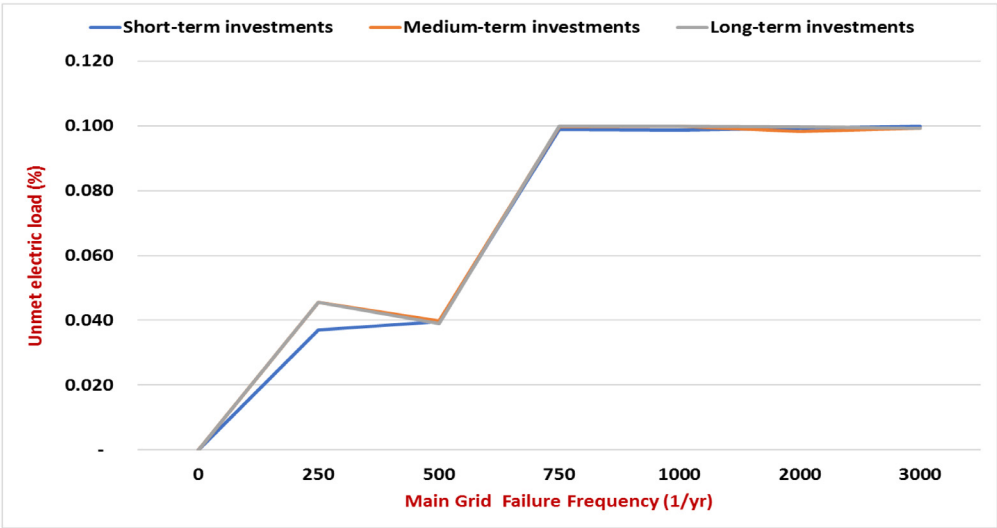


Fig. 13. Unmet electric load for various MGFFs per the various project investment plans.

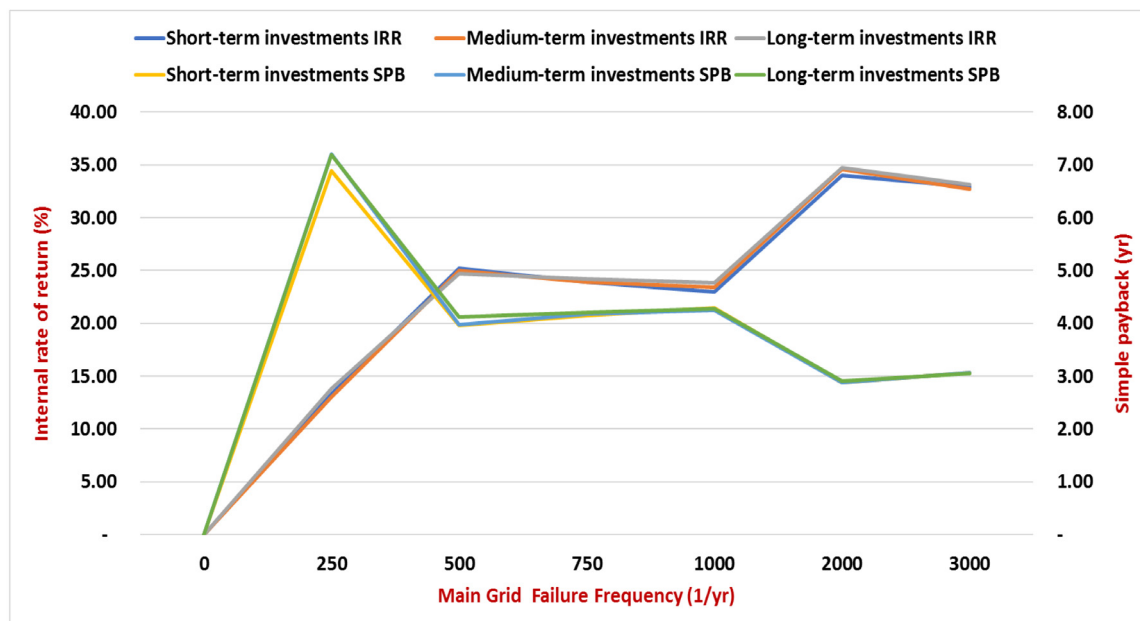


Fig. 14. Internal rate of return (IRR%) and Simple payback (SPB) for various MGFFs per the various project investment plans.

- The investment plan's duration had no impact on the proposed solar PV/battery system size, renewable energy percentage, CO₂ emissions, or unmet electric load.
- The proportion of excess electricity from the optimal system for different MGFFs was obtained, ranging from 0% for zero outage to 13.9% for MGFF 3000. Changing the investment term had just a little influence on the amount of excess electricity produced.
- The CO₂ Emissions from an ideal system for various MGFFs are presented. According to the data, the grid (base scenario) emits the greatest CO₂ (47,289 kg/year) but using hybrid systems connected to the grid reduces carbon dioxide emissions by a whopping 13,496.0 kgCO₂/year.
- The study reveals grid has the least unmet electric load (basic scenario) (with zero percent). However, the rate of increase in all the systems has been slow, with a maximum percentage of 0.1 percent.
- The research calculates the variance of the IRR and SPB for various project investment plans and MGFFs. The maximum IRR and shortest simple payback time are found for GMFFs of 2000 and 3000. Also, the change in investment plans had only a little influence on both IRR and SPB.

The technique used in this research may be applied to other case studies (other sites). Additionally, the observed findings might be approximately extended to places with similar climatic circumstances (solar radiation and ambient temperature).

CRedit authorship contribution statement

Shimaa Barakat: Methodology, Software, Writing – original draft, Writing – review. **A. Emam:** Validation, Formal analysis, Review. **M.M. Samy:** Conceptualization, Formal analysis, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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